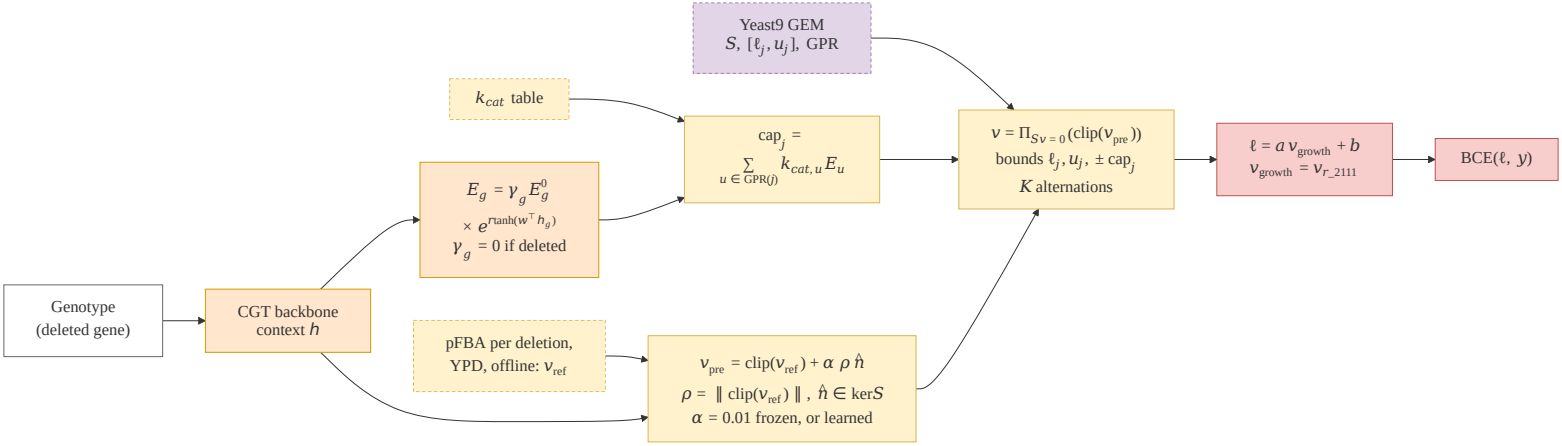

Yeast9 GEM
S, $[\ell_j, u_j]$, GPR
k_{cat} table
Genotype
(deleted gene)
CGT backbone
context h
$E_g = \gamma_g E_g^0$
$\times\ e^{r \tanh(w^\top h_g)}$
$\gamma_g = 0$ if deleted
$\text{cap}_j = \sum_{u \in \text{GPR}(j)} k_{cat,u} E_u$
pFBA per deletion,
YPD, offline: v_{ref}
$v_{\text{pre}} = \text{clip}(v_{\text{ref}}) + \alpha\ \rho\ \hat{n}$
$\rho = \| \text{clip}(v_{\text{ref}}) \|$, $\hat{n} \in \ker S$
$\alpha = 0.01$ frozen, or learned
$v = \Pi_{Sv=0}(\text{clip}(v_{\text{pre}}))$
bounds $\ell_j, u_j, \pm \text{cap}_j$
K alternations
$\ell = a\, v_{\text{growth}} + b$
$v_{\text{growth}} = v_{r_2111}$
BCE(ℓ, y)